

Draw the Line Somewhere Else

84-355 Democracy's Data | The modifiable areal unit problem, on one county's census blocks

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Everything here runs as given. **You will not write any code.** Your job is to read what comes out and say what it means.

Read `areal-units-brief.Rmd` first. This lab runs the same numbered steps with the code exposed. The chunks that *compute* are shown; the chunks that only *draw* are folded away, because their job is pictures, not arithmetic.

The claim under examination

"Fulton County, Georgia is 44.8% Black."

True. Also a fact about a shape somebody drew. This lab holds the census fixed — one day, one count, 1,066,710 people — and changes only the shape.

Two effects, and the whole lab depends on keeping them apart:

- **Scale effect** — same points, bigger boxes. Blocks to block groups to tracts to counties.
- **Zoning effect** — same number of boxes, same size boxes, different shapes. This is what redistricting is.

Step 1 — The same people, all the way down

```
c(ga_blocks = FN("ga_blocks"),
  ga_pop    = FN("ga_pop"),
  ful_blocks = FN("ful_blocks"),
  ful_blocks_with_people = FN("ful_blocks_pop"),
  ful_pop   = FN("ful_pop"),
  ful_black = FN("ful_black"))
```

#>	ga_blocks	ga_pop	ful_blocks
#>	232717	10711908	11748
#>	ful_blocks_with_people	ful_pop	ful_black
#>	8674	1066710	477624

```
c(ful_black_pct = FN("ful_black_pct"), ful_black_vap_pct = FN("ful_bvap_pct"))
```

```
#>      ful_black_pct ful_black_vap_pct
#>           44.78           43.51
```

1,066,710 people on 11,748 blocks, of which 8,674 have anyone living on them. “Black” here means **any part Black** — any of the 32 of the census’s 63 race combinations that include Black or African American, which is the definition voting-rights litigation uses.

Every statistic below is **population-weighted**, so a block holding three people does not count as much as a tract holding six thousand.

Step 2 — Census geography is a string, not a map

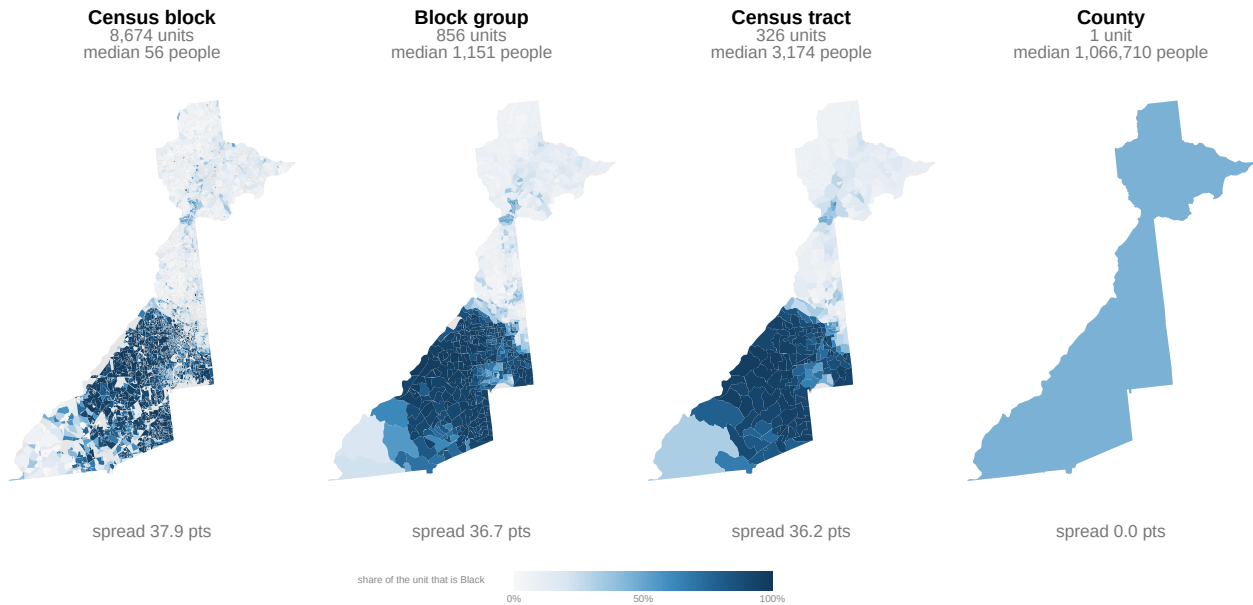
```
eg <- mv[mv$lev == "b", ]
eg <- eg$unit[which.max(eg$pop)]      # the most populous Fulton block
c(block = eg, block_group = substr(eg, 1, 12), tract = substr(eg, 1, 11),
  county = substr(eg, 1, 5), state = substr(eg, 1, 2))
```

```
#>      block      block_group      tract      county
#> "131210007002000" "131210007002" "13121000700" "13121"
#>      state
#>      "13"
```

Climbing the census hierarchy is `substr()`. No overlay, no intersection, no interior points, no assumption about how people are spread inside a polygon. So nothing that changes in Steps 3–5 can be blamed on a geometric approximation: there isn’t one.

GEODs are **character**. Read that block as a number and it becomes 1.312101e+14; read a tract as a number and it silently loses its leading zero. Every read in this document that touches an identifier pins the column.

Step 3 — The scale effect, seen



8,674 shapes on the left, one shape on the right, **exactly the same 1,066,710 people**.

Step 4 — The scale effect, measured

```
lf[, c("level", "units", "median_pop", "mean_share", "sd_share",  
      "p10_share", "p90_share", "units_majority")]
```

Level	Units	Median pop	Mean share	Sd share	P10 share	P90 share	Units majority
Census block	8674	56	44.77543	37.92444	4.140127	97.41379	4129
Block group	856	1151	44.77543	36.69931	6.038201	95.45859	361
Census tract	326	3174	44.77543	36.17960	7.184579	95.15784	135
County	1	1066710	44.77543	0.00000	44.775431	44.77543	0

The **mean is constant at 44.78% on every rung** — it has to be, the same people are being averaged. That is the control. Everything else moves:

```
c(sd_blocks = LF("Census block", "sd_share"),  
  sd_tracts = LF("Census tract", "sd_share"),  
  sd_county = LF("County", "sd_share"),  
  p10_blocks = LF("Census block", "p10_share"),  
  p10_tracts = LF("Census tract", "p10_share"),  
  p90_blocks = LF("Census block", "p90_share"),  
  p90_tracts = LF("Census tract", "p90_share"))
```

```
#> sd_blocks sd_tracts sd_county p10_blocks p10_tracts p90_blocks p90_tracts  
#> 37.924444 36.179603 0.000000 4.140127 7.184579 97.413793 95.157838
```

Statewide the same climb goes further, because Georgia contains between-county variation that Fulton cannot:

```
ls_ga[, c("level", "units", "sd_share", "max_share", "units_majority",
         "pct_black_in_majority", "corr_child")]
```

Level	Units	Sd share	Max share	Units majority	Pct black in majority	Corr child
Census block	165333	31.76072	100.00000	39309	66.33016	0.0882043
Block group	7433	28.40355	99.58621	2070	58.68384	0.0968917
Census tract	2784	27.07374	98.72674	723	55.44737	0.0801182
County	159	17.69965	72.69981	22	34.82968	-0.1467985

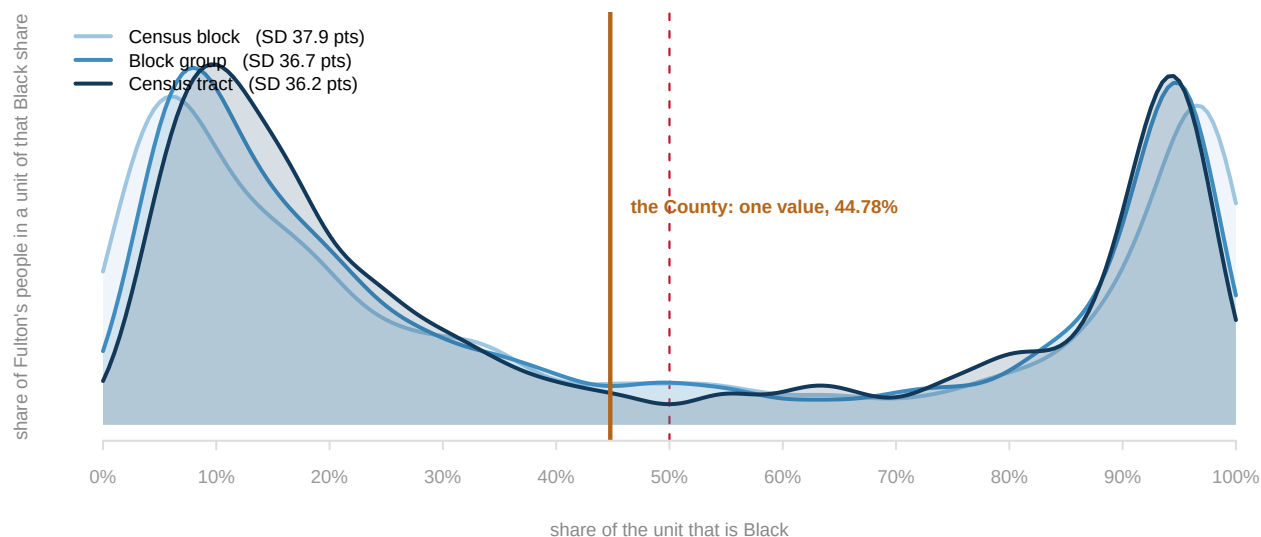
Two results in that table are worth stopping for.

No county in Georgia is as Black as thousands of its blocks are: 100.0% falls to 72.7%. And the **share of Black Georgians who live in a majority-Black unit** falls from 66.3% (blocks) to 55.4% (tracts) to 34.8% (counties). Nobody moved.

The correlation with the under-18 share changes sign — 0.088 across blocks, -0.147 across counties. A correlation is a fact about people *and* the boxes you sorted them into.

Step 5 — Watch the distribution collapse

Same people, four groupings, common bandwidth of 3 points



Two humps, and they stay two humps. Fulton is segregated at a spatial scale *larger than a census tract*, so climbing three rungs barely touches it. The textbook version of the scale effect promises smooth collapse; this is a counterexample, and the reason is worth holding on to: **how much the scale effect bites depends on how big the real spatial structure is compared with the unit.**

Step 6 — The zoning effect: same size units, different shapes

Fulton elects a Board of Commissioners from 6 districts. So take the same blocks and cut them into 6 units of 177,785 people each — 360 different ways, from two rules, each rotated through 180 one-degree steps.

Neither rule can see race.

```
c(plans = nrow(sm), units_per_plan = FN("K"), people_per_unit = FN("quota"),  
  worst_population_deviation_pct = max(sm$max_dev_pct))
```

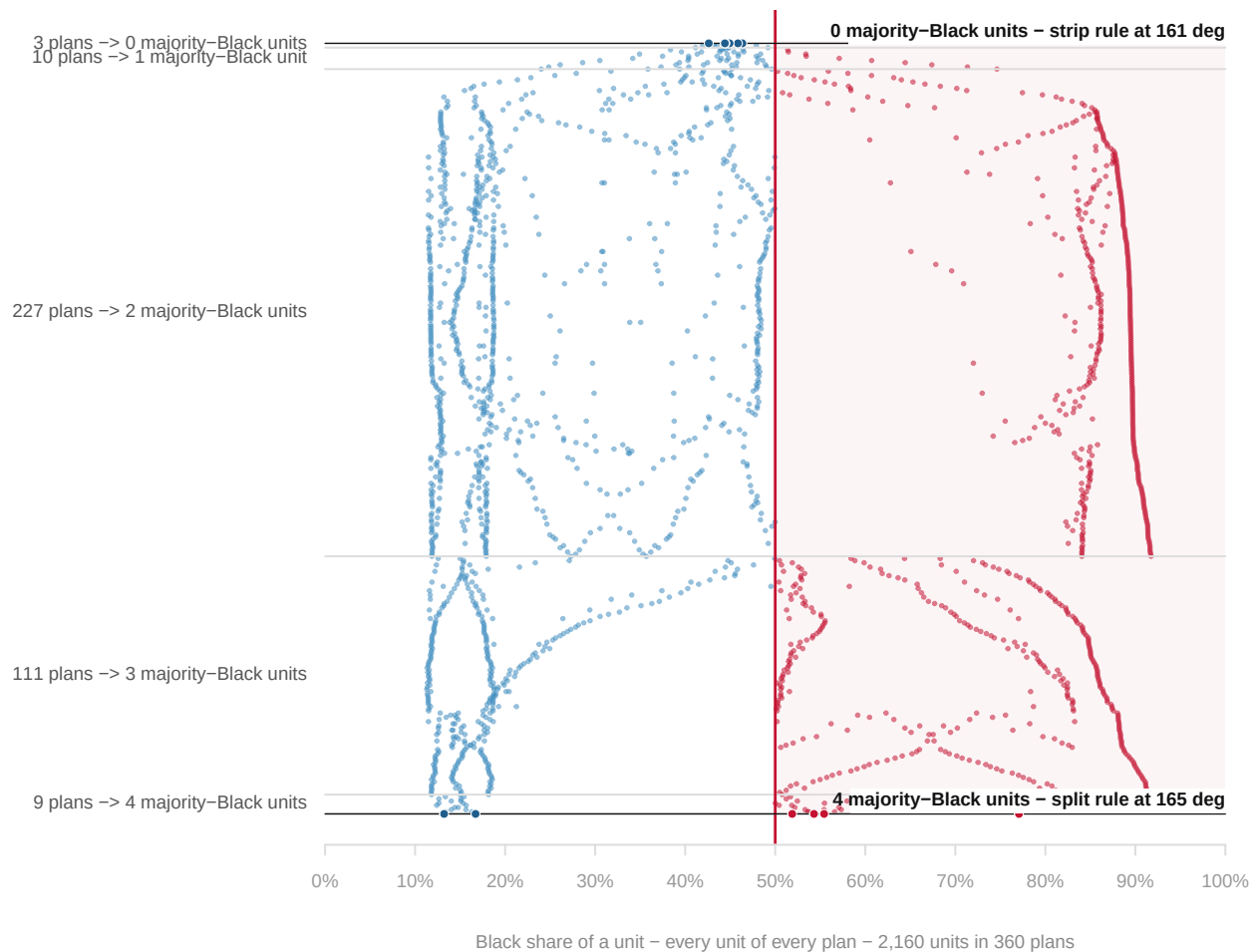
```
#>           plans           units_per_plan  
#>          360.000             6.000  
#>  people_per_unit worst_population_deviation_pct  
#>        177785.000             0.706
```

```
table(family = sm$family, majority_Black_units = sm$n_majority)
```

```
#>      majority_Black_units  
#> family    0    1    2    3    4  
#>  split    0    0 120  51    9  
#>  strip    3   10 107  60    0
```

0 to 4. 3 plans put no unit over 50%. 9 put four over. 227 land on exactly two.

360 equal-population partitions of the same 11,748 blocks



Step 7 — The two extremes, unit by unit

```
z <- pu[pu$plan %in% c("fewest", "most"),
      c("plan", "family", "angle", "unit", "pop", "black", "share", "bvap")]
z
```

Plan	Family	Angle	Unit	Pop	Black	Share	Bvap
fewest	strip	161	1	177848	82443	46.36	44.77
fewest	strip	161	2	178582	81894	45.86	43.96
fewest	strip	161	3	177793	79883	44.93	43.77
fewest	strip	161	4	177788	79057	44.47	44.23
fewest	strip	161	5	177707	78920	44.41	43.50
fewest	strip	161	6	176992	75427	42.62	40.98
most	split	165	1	177998	137188	77.07	76.48
most	split	165	2	177718	98510	55.43	54.59
most	split	165	3	177806	96558	54.31	51.22
most	split	165	4	177590	92142	51.88	49.01
most	split	165	5	177826	29724	16.72	16.31
most	split	165	6	177772	23502	13.22	12.83

The **strip** rule at 161 degrees puts all 6 units between 42.6% and 46.4% — every one of them within a couple of points of the county's own 44.78%, and **none of them a majority**. That is *cracking*, arrived at with no intent at all: the slabs run across the grain of the county, so each picks up part of the Black south-west and part of the white north.

The **split** rule at 165 degrees cuts with the grain instead: 13.2% to 77.1%, and 4 of 6 majority-Black.

A ruler and a protractor did this.

Step 8 — But are they districts? The contiguity test

```
c(plans = FN("n_plans"),
  strictly_contiguous_plans = FN("plans_strict_contig"),
  effectively_contiguous_plans = FN("plans_contig"),
  threshold_pct = FN("contig_thr"))
```

```
#>               plans  strictly_contiguous_plans
#>                360                        0
#> effectively_contiguous_plans  threshold_pct
#>                172                        99
```

```
table(effectively_contiguous = sm$largest_piece_pct >= FN("contig_thr"),
      majority_Black_units = sm$n_majority)
```

```
#>               majority_Black_units
#> effectively_contiguous  0  1  2  3  4
#>                FALSE  3 10 95 71  9
#>                TRUE   0  0 132 40  0
```

Strict contiguity — every block of a unit reachable from every other — is satisfied by **0 plans out of 360**, and the reason is not inkblot shapes. Georgia’s block layer gives the Chattahoochee’s bed, the interstate right-of-way and the airport apron blocks of their own, and they share no boundary with the land across them. Any cut anywhere strands a few near-empty blocks.

So the criterion with meaning is whether a unit’s **people** form one connected mass. Requiring 99% of every unit’s population in a single piece keeps 172 plans — and among them the answer still runs **2 to 3** majority-Black units. The most extreme plans do not survive; the ambiguity does. On a 6-seat board that is one seat, a sixth of the government, decided by which respectable rule you used.

```
pu[pu$plan %in% c("fewest_contig", "most_contig"),
  c("plan", "family", "angle", "unit", "pop", "share", "bvap", "pct_largest")]
```

Plan	Family	Angle	Unit	Pop	Share	Bvap	Pct largest
fewest_contig	split	14	1	178135	85.74	84.90	100.00
fewest_contig	split	14	2	177636	85.53	85.21	99.92
fewest_contig	split	14	3	177644	44.62	43.11	100.00
fewest_contig	split	14	4	177756	22.45	21.74	99.96
fewest_contig	split	14	5	177951	17.39	17.06	99.99
fewest_contig	split	14	6	177588	12.86	12.27	99.99
most_contig	strip	86	1	177793	88.15	87.33	100.00
most_contig	strip	86	2	176990	83.27	82.50	100.00
most_contig	strip	86	3	178516	50.21	46.98	100.00
most_contig	strip	86	4	177542	18.69	18.10	99.94
most_contig	strip	86	5	177887	17.01	17.54	99.98
most_contig	strip	86	6	177982	11.49	11.04	100.00

Step 9 — What you can now say

You can say that a unit’s Black share is a joint fact about the population and the unit, and give the size of the effect: 37.9 points of spread across blocks, 0.0 across the county.

You can say that aggregation destroys tails, and by how much: the share of Black Georgians in a majority-Black unit falls from 66.3% to 34.8% between blocks and counties, with nobody moving.

You can say that a correlation is a property of the grouping, and point to one that changes sign.

You can say that the zoning effect is real and large: 6 equal units, 360 race-blind ways of drawing them, 0–4 majority-Black units, still 2–3 once ill-shaped plans go.

You cannot say that any of these is the plan Fulton should have. Nothing here optimises anything. **The point is not that one of the 360 is right; it is that “what is the Black share of a district here?” has 360 answers before anyone has argued about anything.**

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — A different county. `data/ga_block_race.csv` holds all 232,717 Georgia blocks. Change the five-character county GEOID from Fulton’s 13121 to another county and rebuild the scale ladder. Chatham

(13051), Dougherty (13095), Forsyth (13117) and Clayton (13063) are usefully different from each other. Does the scale effect bite harder or softer than in Fulton, and what does that tell you about the spatial scale of segregation there?

Option B — A different definition of “Black.” The ladder uses `black_any`. Rebuild it with `black_alone` and report what moves and what does not. Which statistic is most sensitive to the definition — the mean, the spread, or the count of majority-Black units? Which choice would you defend in court?

Option C — A different threshold. Fifty percent is a legal convention, not a natural constant. Using `data/zoning_sweep.csv`, recount how many units of each plan clear 40%, 45%, 55% and 60%, and do it with `bvap` (voting-age) as well as `share` (total population). At which threshold does the choice of plan stop mattering — and why is that the interesting number?

Option D — Your own question. 360 partitions, 232,717 blocks, four rungs of census geography.

What this lab cannot tell you

- **Whether any real plan was drawn to do this.** Nothing here has intent. The lab shows the zoning effect is *available*, not that it was used.
- **The true range of possible outcomes.** Two rule families are not the space of all equal-population partitions. 0–4 is a lower bound on the ambiguity, not the answer.
- **Anything about voters.** These are residents. Fulton is 44.78% Black by population and 43.51% by voting-age population, and neither is turnout.
- **Anything about individuals.** Aggregating race to areas and reasoning down to people is the **ecological fallacy**; see `levels-of-aggregation`, and see `rpv` for the one place in this course where the inference can be checked against an answer key.
- **How much of the block counts is noise.** The 2020 census injects differential-privacy noise, and it matters more at block level than at county level — which is itself a scale effect.

On the discussion board

By Friday, 11:59 p.m.

- Step 6 shows two rules that cannot see race producing 0 and 4 majority-Black units. What, if anything, does “the algorithm is race-blind” establish?
- Step 4 shows a correlation changing sign between blocks and counties. If you read a study reporting a correlation across counties, what is the one question you now have to ask?
- Section 2 of the Voting Rights Act asks whether a minority group is “sufficiently large and geographically compact” to constitute a majority in a district. After this lab, is that a question about people, about maps, or about both — and who should get to answer it?

Where the data came from

- **Race by census block** — U.S. Census Bureau, *2020 Census Redistricting Data (P.L. 94-171) Summary File*, legacy format, Georgia; tables P1 (total population by race) and P3 (voting-age population by race). This is the file the Bureau is required by law to deliver to every state legislature so it can draw districts. https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Georgia/ga2020.pl.zip

- **Block, block group and tract boundaries** — Census TIGER/Line 2020, Georgia. https://www2.census.gov/geo/tiger/TIGER2020/BG/tl_2020_13_bg.zip and https://www2.census.gov/geo/tiger/TIGER2020/TRACT/tl_2020_13_tract.zip; blocks (`tl_2020_13_tabblock20`) are shared with `ga-precinct-returns`.
- **Interior points** — the Bureau’s own `INTPTLAT20/INTPTLON20`, the same idea as `assignPolys` in Cervas, *R-Functions*. <https://github.com/jcervas/R-Functions/tree/main/assignPolys>
- **The concept** — Openshaw, S. (1984), *The Modifiable Areal Unit Problem*, CATMOG 38; Robinson, W. S. (1950), “Ecological Correlations and the Behavior of Individuals,” *American Sociological Review* 15(3).

Builds are in `data/`: `build-block-race.R` reads the P.L. 94-171 file into `ga_block_race.csv` (232,717 blocks, 10,711,908 people); `build-areal-units.R` produces the ladders, the maps and the 360 partitions. Both record their source URLs, fetch date and row counts, and both stop if a GEOID is not the width it should be.